

CHAPTER OUTLINE

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KEY POINTS

- Latin America is a heterogeneous region with variable health expenditures and prevalence of kidney disease.
- Mesoamerican nephropathy is a tubulointerstitial kidney disease of unclear cause, with a high prevalence in agricultural workers along the coast of central America.
- Risk factors for Mesoamerican nephropathy may include heat stress, pesticides, heavy metals, and other environmental toxins.
- Kidney involvement in dengue fever varies from acute tubular injury and immune complex glomerulonephritis to thrombotic microangiopathy.
- Leptospirosis affects the kidneys in more than 50% of cases and is characterized primarily by acute interstitial nephritis.

INTRODUCTION

Some 200 years after achieving political independence from European colonizers, Latin America became an area of rapid growth and great economic opportunity. Despite the unifying term, this region is very heterogeneous, consisting of 20 different countries with different languages and degrees of economic as well as social development. Even within a single country such as Brazil, different regions may have diverse levels of development and medical needs.

The total population of Latin America in 2017 was estimated at 645 million people.¹ Brazil, Mexico, Chile, Colombia, and Peru represent more than 70% of the population and account for 75% of its gross domestic product (GDP).¹ Poverty and extreme poverty are present in 29% and 11% of the population, respectively, with a predominance in rural areas.² Average life expectancy in Latin America is 75.7 years; it has risen from 69.7 years in 1995, about 8.6% in 20 years (Fig. 76.1),³ although differences of more than 10 years can be seen between countries. Similar differences are observed in the infant mortality rate, ranging from 6.9 deaths/1000 live births

in Chile to 50 deaths/1000 live births in Bolivia (average, 16 deaths/1000 live births).² Variability in other outcomes, such as maternal mortality, malnutrition, and the lack of access to safe water and sanitation, tend to be obscured by regional averages.² A high prevalence of certain infectious diseases, such as tuberculosis, malaria, dengue fever, and cholera, is also characteristic. Total health expenditures in Latin America rose to 7.3% of the GDP in 2014 from 6.5% in 2008,³ although this figure is still below that of the United States (16.6%) and Europe (9%), and almost half of Latin America's population lacks health insurance (46%; Fig. 76.2).^{2,4}

On the basis of its geographic location, diverse ethnic backgrounds, and socioeconomic status, Latin America is the setting for a number of unique diseases that may affect kidney function; these are reviewed in this chapter. Infections such as dengue fever, malaria, yellow fever, leptospirosis, schistosomiasis, and leprosy are prevalent, and nephrotoxicity from the venom of spiders, caterpillars, or snakes has become more frequent with changes in the environment. In addition, chronic kidney disease (CKD) has become a major health concern with the epidemic of Western metabolic syndrome and unique obscure cases of progressive kidney disease in

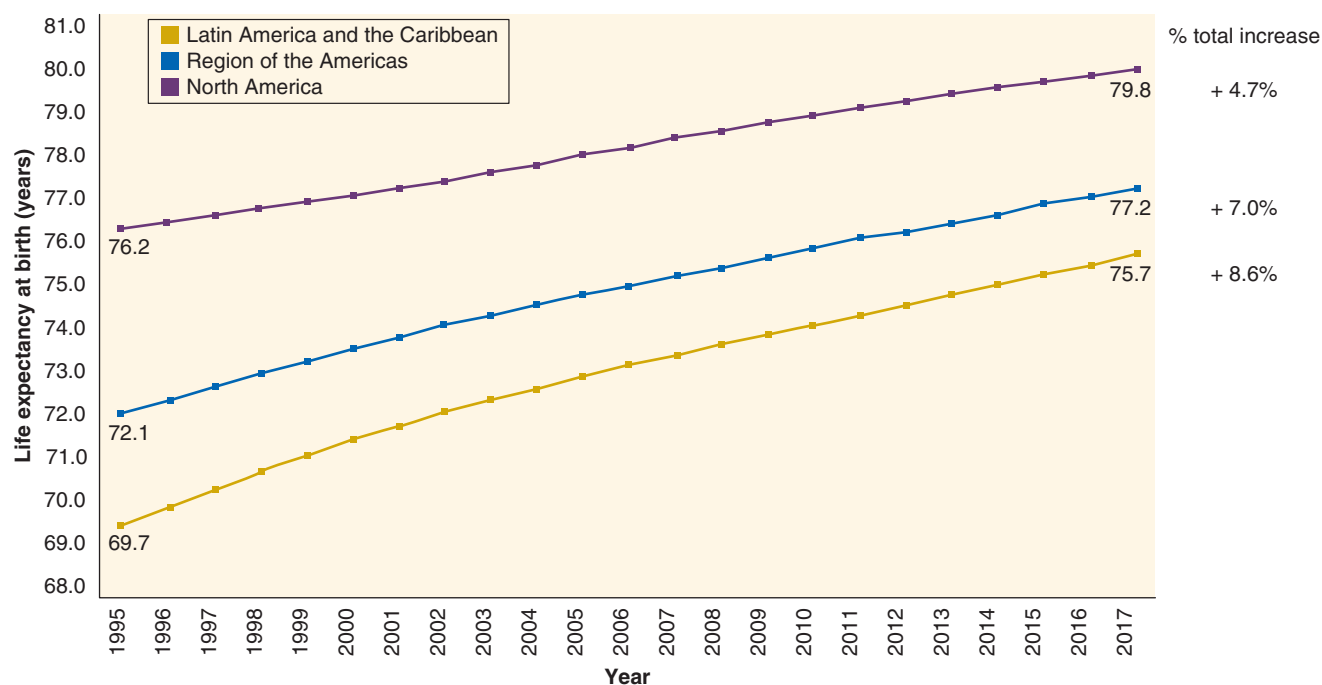


Fig. 76.1 Life expectancy at birth in region of the Americas, North America, Latin America, and the Caribbean, 1995–2017. (Modified from Health in the Americas, 2017 Edition. Summary: Regional Outlook and Country Profiles. <http://iris.paho.org/xmlui/handle/123456789/34321?show=full>.)

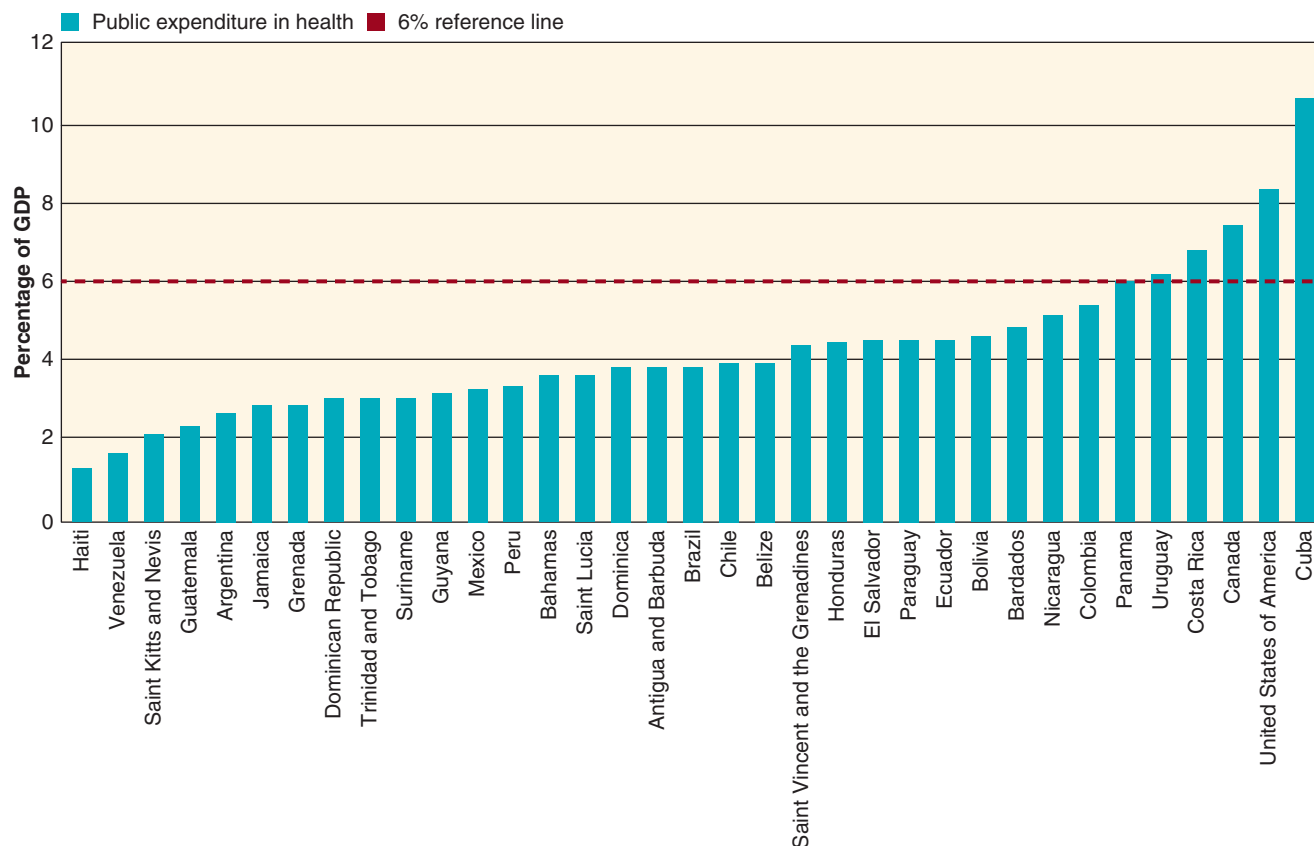


Fig. 76.2 Public health expenditure as a percentage of gross domestic product (GDP), 2014. (Modified from Health in the Americas, 2017 Edition. Summary: Regional Outlook and Country Profiles. <http://iris.paho.org/xmlui/handle/123456789/34321?show=full>.)

sugar cane agricultural workers. This chapter attempts to summarize the current status of the care of patients with kidney disease in Latin America and describe some leading priorities necessary for the improvement and progress of kidney health in the region.

TROPICAL DISEASES AND KIDNEY INJURY

DENGUE FEVER

Dengue fever (DF) is a mosquito-borne infection that causes a severe flu-like illness and, sometimes, a potentially lethal complication called *dengue hemorrhagic fever* (DHF). The main dengue vector is the female of the *Aedes aegypti* mosquito. Male mosquitoes do not transmit the disease because they feed on plant juices. DF and DHF are primarily diseases of tropical and subtropical areas, and the four different dengue serotypes are maintained in a cycle that involves humans and the *Aedes* mosquito. The global incidence of DF has grown dramatically in recent decades.

Approximately half of the world's population lives in areas potentially at risk for DF, and 50 to 100 million cases are estimated to occur annually.^{5,6} Brazil has the highest absolute number of cases and the highest incidence of the disease in Latin America.⁵ The principal method for prevention of DF virus transmission is mosquito control. The high incidence of DF in Latin America has been ascribed to global warming and ecosystem modifications, facilitating the dissemination of pathogens and vectors of the disease. In addition, demographic factors, including uncontrolled migration of people and unplanned urbanization, have facilitated the increase in the occurrence of DF in urban areas and large cities.

DF is caused by one of four closely related, but antigenically distinct, virus serotypes—DEN-1, DEN-2, DEN-3, and DEN-4—of the genus *Flavivirus*. Infection with one of these serotypes provides immunity to only that serotype for life; therefore, persons living in a DF-endemic area can have more than one DF infection during their lifetimes.

Dengue virus infection may manifest clinically as an undifferentiated fever—DF, DHF, or dengue shock syndrome (DSS). DHF usually occurs in patients who have previously been infected with other serotypes of dengue virus ("secondary infection"), although it may follow primary infections, particularly in infants.⁶ Classic DF is an acute febrile illness accompanied by headache, retroorbital pain, and muscle and joint pains that lasts approximately 5 days but that has no specific physical findings except a maculopapular rash in about 50% of cases. DHF is a severe form of the disease and manifests with fever, hemorrhagic phenomena, thrombocytopenia, and evidence of plasma leakage (high hematocrit, pleural effusion, ascites, and hypoalbuminemia). Malaria, leptospirosis, hantavirus infection, typhoid fever, human immunodeficiency virus (HIV) infection, enterovirus infection, influenza, and sepsis must be ruled out in the differential diagnosis. The diagnosis of acute dengue virus infection is mainly a clinical one, but serologic tests are available in specialized reference laboratories, including hemagglutination inhibition (HI) assay (considered the gold standard), immunoglobulin (Ig) G or IgM enzyme immunoassays, and reverse transcription polymerase chain reaction (RT-PCR) analysis for dengue virus RNA.

KIDNEY INVOLVEMENT IN DENGUE FEVER

Renal injury, comprised of acute tubular injury, glomerulonephritis, or hemolytic-uremic syndrome, has been reported in patients with DF. The clinical presentation is variable, ranging from an elevation in creatinine level, proteinuria, and active urinary sediment to thrombotic microangiopathy. The possible pathogenesis of renal involvement includes ischemic or hemoglobin-associated tubular injury, glomerular injury, secondary immune complex deposition, and thrombotic microangiopathy.⁷ Lima and colleagues have described a case of DHF-induced acute kidney injury (AKI), in which renal injury occurred without hemodynamic instability, hemolysis, rhabdomyolysis, or use of nephrotoxic drugs.⁵

The prevalence of AKI in dengue infection is variable, ranging from 0.3% to as high as 15%.⁷ In a series from Thailand, 51 deaths were reported among 6154 cases of DHF, with an overall incidence of AKI of 0.3%, but an incidence of 33.3% among the fatal cases. Kidney biopsies performed in patients with DHF and proteinuria, hematuria, or both revealed glomerular changes characterized by hypertrophy and hyperplasia of mesangial and endothelial cells, the presence of monocyte-like cells in some of the glomerular capillary lumens, and focal thickening of the glomerular basement membrane. Immune complexes (IgG, IgM or both, and C3) were found in glomeruli and arterioles in biopsy specimens from 10 cases that were collected 2 weeks after the onset of symptoms. Dense spherical particles were found in the 12 cases in which electron microscopy was carried out. The researchers in this series hypothesized that the particles might be nucleocapsid cores of dengue virions.⁸

Experimental studies with inoculation of the virus in mice have suggested that the dengue virus can induce glomerulopathy. In one study in which dengue virus type 2 was inoculated into mice, diffuse proliferative glomerular injury was seen 14 days after the inoculation and, in another study, enlarged glomerular volume, increased endocapillary and mesangial cellularity, and glomerular IgM deposition were observed 48 hours after virus inoculation.⁹ In a case report from Brazil, the presence of proteinuria, hematuria, generalized edema, and hypertension associated with low levels of C3 and C4 suggested an immune-mediated, acute glomerular injury.⁷

There is no specific treatment for DF, so management consists of supportive care, including controlling fever (acetaminophen), preventing dehydration (oral [PO] or intravenous [IV] fluids), and managing complications such as bleeding. Aspirin and nonsteroidal antiinflammatory drugs (NSAIDs) should be avoided. There are no vaccines available, and the only mode of prevention is to protect against the mosquito bite by covering exposed skin areas and using an effective mosquito repellent, such as *N,N*-diethyl-metaltoluamide (DEET).

YELLOW FEVER

Yellow fever (YF) is a viral hemorrhagic fever transmitted by infected mosquitoes. There are three types of transmission cycle—sylvatic, intermediate, and urban. All three cycles exist in Africa, but in South America, only sylvatic and urban yellow fever occur.¹⁰ Sylvatic (or jungle) YF occurs in tropical rainforests, where monkeys, infected by sylvatic mosquitoes,

pass the virus to other mosquitoes that feed on them; these mosquitoes, in turn, bite and thus infect humans entering the rainforest. This type of transmission produces sporadic cases, most of which affect young men working in the forest (e.g., loggers). The intermediate cycle of YF transmission, which occurs in humid or semihumid savannahs of Africa, can produce small-scale epidemics in rural villages. Semidomestic mosquitoes infect monkey and human hosts, and increased contact between human and infected mosquito leads to disease. This is the most common type of outbreak seen lately in Africa. Urban yellow fever results in large, explosive epidemics when travelers from rural areas introduce the virus into areas with a high human population density. Domestic mosquitoes, most notably *Aedes aegypti*, carry the virus from person to person. These outbreaks tend to spread outward from one source to cover a wide area.

YF transmission in the Americas continues to follow the jungle cycle. There are 14 YF-endemic countries in the region. However, only five of them (Bolivia, Brazil, Colombia, Ecuador, and Peru) reported cases of yellow fever between 2011 and 2016. As of March 2017, the Pan American Health Organization reported that Brazil, Colombia, Peru, and the plurinational state of Bolivia had suspected and confirmed yellow fever cases.¹⁰ In Brazil, since the outbreak of December 2016, 1368 cases were reported—326 confirmed, 125 excluded, and 916 suspected cases remain under investigation—including 220 deaths (109 confirmed, 6 discarded, and 105 under investigation). The case fatality rate (CFR) is 33% among confirmed cases and 11% among suspected cases.¹⁰

To avoid the reurbanization of this disease *A. aegypti* vector control measures are critical, mainly in cities and in localities bordering affected areas. Such measures also facilitate the prevention of dengue outbreaks. One of the most important mechanisms for preventing YF is vaccination. The live attenuated vaccine against YF confers protective immunity in about 90% of individuals within 10 days of receiving the 0.5-mL subcutaneous dose and in nearly 100% of individuals within 3 to 4 weeks after vaccination. A booster dose is recommended every 10 years. The World Health Organization (WHO) recommends vaccination for travelers to YF-endemic areas of Africa and South America, as well as for residents of those areas. The vaccine is contraindicated in persons who have known egg allergy or are immunosuppressed (e.g., transplant recipients). YF causes epidemics that can affect 20% of the population. When epidemics occur in unvaccinated populations, case fatality rates may exceed 50%. Clinical presentation is characterized by three stages—infection, remission, and intoxication. The period of infection consists of viremia with nonspecific symptoms and signs including fever, malaise, headache, joint pain, nausea, and vomiting. This is followed by a period of remission, with resolution of symptoms for up to 2 days. The subsequent period of intoxication is characterized by hepatic dysfunction, AKI, coagulopathy (low platelets), and shock. Diagnosis is confirmed by serologic testing such as IgM enzyme immunoassays or RT-PCR analysis for the YF virus genome.

KIDNEY INVOLVEMENT IN YELLOW FEVER

Renal complications usually occur in the severe form of the disease between the fifth and seventh days, with reduction in urine volume and albuminuria. The mechanism of kidney involvement is poorly understood, although multiple causes

are possible, including ischemic tubular injury due to shock, rhabdomyolysis, glomerulonephritis, and acute interstitial nephritis.^{11,12} In experimental kidney injury, during the first 24 hours, kidney dysfunction appears to be related to hypovolemia and, later, oliguria, metabolic acidosis, albuminuria, and cylindruria occur. Viral antigen has been found in glomeruli and tubules. No treatment beyond supportive care is currently available.¹⁰

MALARIA

Malaria is caused by *Plasmodium* parasites, which are carried by mosquitoes and transmitted to people through their bites. In Latin America and the Caribbean, 75% of malaria infections are caused by *Plasmodium vivax* and are rarely fatal; 25% are caused by the more lethal *Plasmodium falciparum*, the dominant malaria parasite in Africa.

Between 2000 and 2015, the number of cases of malaria in Latin America declined by 62%.¹³ Of the total cases, 77% were reported by Brazil, Peru, and Venezuela. During the same period, malaria-related deaths fell by 76%. At the end of 2015, malaria was still endemic in 21 countries of Latin America.¹³ Venezuela saw a 50% increase in its malaria cases between 2014 and 2015. If this trend continues, it is expected that Venezuela will have the largest number of cases in the region.^{3,13} Despite the achievements, there are still an estimated 108 million people at risk for malaria, of which at least 5.7 million were classified as being at high risk.¹³ The progress so far is largely due to improved treatment of the most deadly form of the disease, along with more effective mosquito control.

The clinical manifestations of malaria vary according to geography and prior immunity. Following the bite of an infected female *Anopheles* mosquito, the inoculated sporozoites go to the liver within 1 hour. Individuals are asymptomatic for 2 to 3 weeks, depending on parasite species, until the erythrocytic stage of the parasite life cycle. The release of merozoites from infected red blood cells when they rupture causes fever, malaise, headache, nausea, vomiting, and myalgias. Severe cases of hyperparasitemia may be associated with shock, acute respiratory distress syndrome (ARDS), AKI, coagulopathy, and hepatic failure. Physical findings include pallor, petechiae, jaundice, hepatomegaly, and splenomegaly. Clinical suspicion should prompt a confirmatory test for the parasite, which may involve light microscopy (visualization of parasites in stained blood samples), a rapid diagnostic test (detecting antigen or antibody), or a molecular technique for detecting parasite genetic material.

KIDNEY INVOLVEMENT IN MALARIA

AKI is one of the most dreaded complications of severe malaria. According to WHO criteria, AKI (serum creatinine level ≥ 3 mg/dL, or ≥ 265 μ mol/L) occurs as a complication of *P. falciparum* malaria in less than 1% of cases, but the mortality rate in these cases may be up to 45%.¹⁴ Malaria-induced AKI is more common in adults than children. Renal involvement varies from mild proteinuria to severe azotemia associated with metabolic acidosis. It may be oliguric or nonoliguric. AKI may occur as a component of multiorgan dysfunction or as a lone complication. AKI is seen mostly in *P. falciparum* infections, but can occasionally occur with *P. vivax* and *P. malariae*. Malarial AKI is predominantly found

in nonimmune adults and older children with falciparum malaria.

Several hypotheses have been proposed to explain the pathogenesis, including mechanical obstruction of glomerular capillaries by infected erythrocytes, immune-mediated glomerular injury, and acute tubular necrosis secondary to shock and/or free hemoglobin tubular toxicity.¹⁵ It is likely that more than one of these mechanisms may occur simultaneously to produce the clinical manifestation. The predominant lesions on biopsy are acute tubular necrosis and mild proliferative glomerulopathy. Patients with these findings tend not to progress to CKD. Excessive fluid administration, oxygen toxicity, and yet unidentified factors may contribute to pulmonary edema, ARDS, multiorgan failure, and death.¹⁴

The management of malaria-induced AKI includes appropriate antimalarials (parenteral artesunate or quinine), fluid and electrolyte management, and early renal replacement therapy (RRT). Currently, high-quality intensive care, early institution of RRT, and avoidance of nephrotoxic drugs are standard practice for the prevention and management of AKI. Preliminary data have suggested that albumin infusion for volume expansion may reduce mortality rates.¹⁶

LEPTOSPIROSIS

Leptospirosis is a zoonosis with protean manifestations caused by the spirochete *Leptospira interrogans*. The genus *Leptospira* consists of two species, *Leptospira interrogans* and *Leptospira biflexa*, only the former of which is known to cause human disease. More than 200 serovars of *L. interrogans* have been identified.

Leptospirosis is distributed worldwide, except for the polar regions. However, most clinical cases occur in the tropics. Leptospirosis presents a greater problem in humid tropical and subtropical areas, where most developing countries are found, than in areas with temperate climates. The magnitude of the problem in tropical and subtropical regions can be largely attributed not only to climatic and environmental conditions, but also to the greater likelihood of contact with a *Leptospira*-contaminated environment caused by, for example, local agricultural practices, as well as poor housing and waste disposal, all of which give rise to many sources of infection.¹⁷

Human leptospirosis is endemic in Latin America and usually reaches epidemic levels after higher rainfall periods with flooding or natural disasters such as hurricanes. In Brazil, 23,574 cases were reported from 1997 through 2008, with a mortality rate of 11.2%, and with similar figures reported from 2009 to 2017.¹⁸ The number of cases in Brazil has steadily declined: 4341 in 2015, 2870 in 2016 and, as of September 2017, 2003.¹⁸ Similarly, the death rate has declined from 334 in 2015 to 234 in 2016 and to 231 in 2017.¹⁹

The natural hosts for the organism are various mammals, including rodents, dogs, pigs, cattle, and horses. Humans are only incidentally infected, typically after exposure to the environment contaminated by animal urine, contaminated water or soil, or infected animal tissue. Portals of entry include cuts and abraded skin, mucous membranes, and conjunctiva. The infection is rarely caused by the ingestion of food contaminated with urine or via aerosols. Controversy exists as to whether *Leptospira* can penetrate intact skin.²⁰

Leptospirosis is a potentially serious but treatable disease. It can provoke a broad range of manifestations, from benign

infection, characterized by nonspecific symptoms, to Weil disease, a severe form of the disease that causes jaundice, hemorrhagic events, and AKI. Its symptoms may mimic those of a number of other unrelated infections, such as influenza, meningitis, hepatitis, dengue, and viral hemorrhagic fever.

Leptospirosis manifests as an abrupt onset of fever, rigors, myalgias, and headache in 75% to 100% of patients after an incubation period of 2 to 26 days (average, 10 days). After the acute septicemic stage, hyperbilirubinemia occurs in 85% of cases.²¹ On physical examination, conjunctival suffusion is an important sign. Most patients have muscle tenderness, splenomegaly, lymphadenopathy, pharyngitis, hepatomegaly, muscle rigidity, abnormal respiratory auscultation, and/or rash.

The most widely used method for diagnosing leptospirosis is the microscopic agglutination test (MAT), which is performed using two blood samples collected 2 weeks apart. The MAT results are considered positive when the antibody titers are fourfold higher than the normal reference value.²² Available in some laboratories, detection of antigens by a PCR assay offers the possibility of earlier diagnosis.

KIDNEY INVOLVEMENT IN LEPTOSPIROSIS

Renal involvement is common in leptospirosis. Clinical manifestations vary from urinary sediment changes to AKI, which is observed in 44% to 67% of patients.^{21,23} Leptospirosis manifesting as AKI is a severe disease, frequently leading to multiorgan failure and death. Leptospirosis-induced AKI typically is nonoliguric and often includes hypokalemia.²²

The renal pathologic findings in leptospirosis are characterized by acute interstitial nephritis that may be associated with acute tubular necrosis.²⁴ The tubular injury may reflect a direct toxic effect of leptospiral compounds on tubular epithelial cells or indirect effects of dehydration, hypovolemia, and ischemia due to ionic wasting defects and the inability to concentrate urine.^{20,21} Vasculitis is also observed in the acute phase of the disease. A cellular infiltrate consisting primarily of mononuclear cells can be diffuse or focused around the glomeruli and venules.²² Studies have suggested that inflammation is most likely a secondary feature following acute epithelial damage and is observed only in patients who survive long enough for this feature to develop.²⁵

Silver and other immunohistochemical staining reveals large numbers of intact leptospirae throughout the tubular basement membrane, among tubular cells, and within the tubular lumens and the interstitium, with associated interstitial nephritis.²⁴ In some cases, organisms are seen in limited numbers within glomeruli, but glomerular changes in general are not remarkable.

Experience with leptospirosis-associated AKI in Thailand has shown that the use of hemodialysis or hemofiltration is associated with lower mortality (0% vs. 10%), shorter recovery time (8 vs. 16 days), and faster reduction in serum levels of bilirubin, urea, and creatinine in comparison with standard peritoneal dialysis.²⁶ There is, however, no data from randomized clinical trials evaluating different methods of RRT in patients with leptospirosis.²⁰

The vast majority of infections with *Leptospira* are self-limiting. Two small randomized, placebo-controlled trials have shown a benefit of antimicrobial therapy. In one of these trials, in which doxycycline (100 mg PO, bid) was compared with placebo, doxycycline shortened the illness

by an average of 2 days and prevented shedding of the organism in the urine. In a second trial, patients with severe leptospirosis who were treated with penicillin had fewer days of fever, more rapid resolution of serum creatinine level elevations, and shorter hospital stays; penicillin therapy also prevented urinary shedding. Therefore, symptomatic patients should receive antimicrobial therapy to shorten the duration of illness and reduce shedding of organisms in the urine. For patients with mild leptospirosis a course of doxycycline, amoxicillin, or azithromycin is recommended. Severe leptospirosis is usually treated with intravenous penicillin (1,500,000 U every 6 hours). Intravenous ceftriaxone (1 g once daily) or cefotaxime (1 g every 6 hours) has efficacy equivalent to that of penicillin. Treatment must be maintained for 7 days.

Doxycycline prophylaxis appears to be protective, reducing morbidity and mortality during outbreaks.²² Control of reservoirs is difficult, and no vaccines are available. China and Brazil, countries where leptospirosis is a major health problem, have completed the sequence of the *L. interrogans* genome. Together with new genetic tools and proteomics, novel insights have been made into the biology of *Leptospira* and the mechanisms used to adapt to host and external environments. Surface-exposed proteins and putative virulence determinants have been identified that may serve as subunit vaccine candidates.

NEPHROTOXICITY DUE TO VENOMS

SPIDERS

Spiders of the genus *Loxosceles* are known colloquially as recluse spiders, violin spiders, fiddleback spiders and, in South America, by the nonspecific name “brown spiders.”

In South America, *Loxosceles* spiders of medical importance are found in Brazil and Chile. The most common species involved in envenomations are *Loxosceles laeta*, *Loxosceles intermedia*, and *Loxosceles gaucho*. *L. laeta* is often considered the most dangerous of the recluse spiders, in part because it is the species that attains the largest body size. Recluses are found mostly inside homes, in basements, in attics, behind bookshelves and dressers, and in cupboards. As their name implies, these spiders prefer dark quiet areas that are rarely disturbed. Outdoors, they are found under objects, such as rocks and the bark of dead trees.

Loxosceles spider bites are the leading cause of spider envenomation and necrotic arachnidism in humans. The venom of the spiders is inoculated to paralyze and digest their prey. “Loxoscelism,” the term used to describe lesions and reactions induced by bites from spiders of the genus *Loxosceles*, causes skin necrosis, rhabdomyolysis, hemolysis, coagulopathy, AKI, and systemic inflammatory response syndrome (SIRS) in humans (Fig. 76.3). Loxoscelism is currently considered a serious public health problem in southern Brazil. There are 3000 reports of *Loxosceles* bites annually, and it constitutes the third leading cause of accidents involving venomous animals in Brazil.²⁷

The venom can trigger a powerful inflammatory response characterized by local and systemic production of cytokines and chemokines, as well as the activation of complement and nitric oxide production, together with inflammatory cell migration and platelet aggregation.²⁸ The venom also has a direct hemolytic effect on red blood cells and damages the membranes of endothelial cells in blood vessel walls.²⁹

KIDNEY INVOLVEMENT IN LOXOSCELISM

Rhabdomyolysis and hemolysis are recognized cofactors involved in the genesis of AKI,²⁹ and there is also evidence of a direct nephrotoxic action. The nephrotoxic effect of



Fig. 76.3 Evolution of skin lesions after brown spider bite.

the *Loxosceles* spider venom has been demonstrated experimentally in mice exposed to whole venom. Light microscopic analysis of kidney biopsy specimens has shown alterations, including hyalinization of proximal and distal tubules, erythrocytes in Bowman's space, glomerular collapse, tubule epithelial cell blebs and vacuoles, interstitial edema, and deposition of eosinophilic material in the tubule lumen.³⁰ Electron microscopic findings have indicated changes, including glomerular epithelial and endothelial cell cytotoxicity and disorders of the basement membrane. Tubule alterations include epithelial cell cytotoxicity with cytoplasmic membrane blebs, mitochondrial changes, increase in smooth endoplasmic reticulum, the presence of autophagosomes, and deposits of amorphous material in the tubules. In the same study, confocal microscopy with antibodies against venom proteins showed direct binding of toxins to renal structures, confirmed by competition assays.³⁰

The treatment of a recluse spider bite involves local wound care, pain management, and tetanus prophylaxis, if indicated. Dapsone may be administered in some cases, both to prevent progression to necrosis and to reduce pain. Antivenoms for the treatment of recluse spider bites are available in Brazil, Mexico, and Peru, although not in the United States. The bites of South American *Loxosceles* species (e.g., *L. gaucho*) are more severe than those of recluse spiders found in the United States.

The results of studies involving animal models have suggested that specific types of antivenom decrease the lesion size and limit systemic illness, even when its administration is delayed. To date, the most extensive use of antivenom treatment has been in Brazil, and the Brazilian Ministry of Health has developed guidelines for its use in patients with extensive cutaneous lesions or severe systemic illness.²⁹ South American recluse antivenoms may help reduce the risk of dermonecrosis as well as systemic envenomation and its severe complications (e.g., hemolysis, AKI, disseminated intravascular coagulation [DIC]). The data regarding the efficacy of antivenom treatment are largely based on animal studies, and the benefit in humans has not been well established.

LONOMIA CATERPILLARS

Caterpillars are the larval stage of moths and butterflies and are found worldwide.³¹ *Lonomia* caterpillars (family, Saturniidae) are found in South America, and their venom affects mainly the coagulation system.³² Accidental poisoning by caterpillars has become increasingly frequent in southern Brazil, partly owing to deforestation and the elimination of natural predators.³¹ Accidental contact with caterpillars' bristles induces allergic and toxic signs and symptoms that range from a mild cutaneous reaction to severe systemic reactions, depending mainly on the number and species of the caterpillar involved. Symptoms include local irritation, urticarial dermatitis, allergy, ocular injuries, osteochondritis, hemorrhage, secondary coagulopathy, and AKI. Hemorrhagic complications, including intracerebral hemorrhage, can result in death.^{22,31}

Caterpillars cluster in large numbers on tree barks and blend in very well (Fig. 76.4). Contact with large numbers of caterpillars can occur, exposing the individual to large dangerous doses of venom. Two species of *Lonomia*, the Brazilian caterpillar *Lonomia obliqua* and the Venezuelan



Fig. 76.4 Cluster of caterpillars on the trunk of a tree. (Courtesy M. Vennegoor.)

caterpillar *Lonomia achelous*, provoke activation of the coagulation cascade through the action of several compounds that are the subject of toxicology research.³¹ In *L. obliqua* bristle extract toxins have been identified, including a factor X activator named Losac and a prothrombin activator named Lopap.³³ Zannin and colleagues have reported a series of 105 patients poisoned by *L. obliqua* and found coagulation factor depletion and enhanced levels of fibrin degradation in keeping with DIC, although platelet counts were typically preserved.³⁴

Mortality in Brazil secondary to these toxins peaked at 20% before falling to less than 2% following the introduction of the anti-*Lonomia* horse serum (in Brazil) in 1995 and educational campaigns.

RENAL INVOLVEMENT IN LONOMIA ACCIDENTS

Since 1989, the southern Brazilian states have reported accidental poisoning by *L. obliqua* in 2067 individuals. AKI developed in 1.9% of patients and, of these, 32% required dialysis, with 10.3% going on to have CKD.³⁵ Our group has reported an event with a successful outcome due to early recognition and treatment with the antivenom serum.³⁶ Even with anti-*Lonomia* serum, however, AKI can still occur.³⁷ Coagulation factors tend to be more profoundly altered in all patients with AKI, in keeping with their having more severe disease. The renal lesion appears to be secondary to massive deposition of fibrin in the glomeruli, leading to ischemia.^{37,38} Some of the toxin components also cause direct toxicity to endothelium and tubular cells. The histology is poorly characterized because the coagulopathy prevents biopsies in the acute phase of the illness.

When traveling to endemic zones in Central and South America, travelers should be aware of poisonous caterpillars and avoid direct contact. Accidental poisoning should be reported immediately to the health authorities nearest the

site where the poisoning took place because they are more likely to have immediate access to anti-*Lonomia* serum. Most countries do not stock this product, so timely administration and supportive measures are critical.

SNAKES

Envenomation resulting from snakebites is a particularly important public health problem in rural areas of tropical and subtropical countries in Africa, Asia, Oceania, and Latin America. One study has estimated that at least 421,000 envenomations and 20,000 deaths occur worldwide from snakebite each year but warns that these figures may be as high as 1,841,000 envenomations and 94,000 deaths. The highest burdens of snakebites are in South Asia, Southeast Asia, and sub-Saharan Africa.³⁹

There are approximately 3000 species of snakes, of which approximately 19% are venomous. Most snakebites occur in tropical regions, where they represent a serious public health burden because of the resulting morbidity and mortality. Latin America is the third most affected area after Asia and Africa. Snakebites are more common in the rainy seasons and are related to the increase in human activity in rural areas. The most affected group is 25- to 49-year-old men. Lower limbs are the most frequently injured sites.

Epidemiologic data on snakebite envenomation in Latin America are scarce. In Brazil, there are 20,000 accidents involving venomous snakes annually, an incidence of 13.5 accidents/100,000 inhabitants, and a mortality rate of about 0.45%. AKI is one of the main complications after snakebite envenomation and it is an important cause of death in affected patients. Snakebite-induced renal injury has been reported with almost all venomous snakes; however, AKI is more frequent with *Vipera russelli* in Asia, as well as *Bothrops* and *Crotalus* spp. in South America.⁴⁰

CROTALUS SNAKES

The South American rattlesnake, which belongs to the Viperidae family, Crotalinae subfamily, *Crotalus* genus, is represented in Brazil by a single species, *Crotalus durissus*, distributed into five subspecies, of which *Crotalus durissus terrificus* and *Crotalus durissus collilineatus* are the most important. *Crotalus* venom effects are multiple; the most important clinical manifestations are neurotoxicity, myotoxicity, nephrotoxicity, and coagulopathy.⁴¹

Crotalus venom induces clinical manifestations that can vary from mild local to severe systemic manifestations.⁴¹ Neurotoxicity may manifest as eyelid ptosis, blurred and/or double vision, ophthalmoplegia, and facial paralysis. Myotoxicity is revealed by generalized myalgia and myoglobinuria due to the rhabdomyolysis. The coagulopathy caused by the thrombin-like enzyme produces blood incoagulability and afibrinogenemia in 40% to 50% of patients, although bleeding is a rare manifestation.⁴¹

Kidney Involvement in *Crotalus* Bites

The prevalence of AKI associated with *Crotalus* envenomation ranges from 10% to 29% and occurs within the first 24 to 48 hours after the bite. Experimental and clinical studies have suggested that the pathogenesis of *Crotalus* venom-induced AKI is likely related to rhabdomyolysis, renal vasoconstriction, and direct tubular cell toxicity.

Crotalus venom and probably crotoxin, the most important component of the venom, have direct and indirect actions on renal cells. Histologic injury usually found in *Crotalus* snakebite victims is acute tubular necrosis, although cases of interstitial nephritis have also been reported.^{42,43} Shock, hypotension, hemolysis, sepsis, or nephrotoxic drugs can contribute to the AKI. In one report, dialysis was necessary in 24% of cases.⁴⁴ Mortality rates reported for AKI after *Crotalus* envenomation range from 8% to 17%.

BOTHROPS SNAKES

In Latin America, the vast majority of snakebite envenomations are inflicted by species of the genus *Bothrops*. *Bothrops asper*, a species widely distributed in southern Mexico, Central America, and the northern areas of South America, is responsible for most cases in these regions.⁴⁵

Snakes of the *Bothrops* genus belong to the Viperidae family and Crotalinae subfamily. There are more than 30 species distributed from southern Mexico to Argentina and Brazil. The most important species are *Bothrops asper* in Central America and *Bothrops atrox*, *Bothrops erythromelas*, *Bothrops neuwiedi*, *Bothrops moojeni*, *Bothrops jararaca*, *Bothrops jararacussu*, and *Bothrops alternatus* in Brazil.⁴¹ These species are found in rural areas and on the outskirts of large cities and prefer humid environments, such as forests, plantation areas, and structures (e.g., barns, silos, and wood deposits), which are thought to contribute to the proliferation of snakes and rodents. These animals have nocturnal or crepuscular habits and an aggressive defensive behavior.⁴¹

Bothrops venom has proteolytic, coagulant, and hemorrhagic actions. Local manifestations at the bite site such as edema, blisters, and necrosis are caused by the venom's proteolytic action.⁴⁵ A direct nephrotoxic action of the venom has also been shown.⁴⁵ Lesions result from the activity of proteases, esterases, hyaluronidases, and phospholipases (phospholipase A2 [PLA2]), release of inflammatory mediators, action of hemorrhagins on the vascular endothelium, and procoagulant action of the venom.^{41,46} The pathogenesis of these pathologic alterations has been investigated in experimental models. The observed effects result mostly from the combined action of Zn⁴⁺-dependent metalloproteinases (snake venom metalloproteinases [SVMPs]) and myotoxic PLA2.⁴⁶ SVMPs are able to degrade extracellular matrix components, such as those present in the basement membrane of microvessels and at the dermal-epidermal junction, whereas myotoxic PLA2 disrupts the integrity of the plasma membrane of skeletal muscle fibers. In addition, a prominent inflammatory reaction of multifactorial origin develops, resulting in a pronounced edema and an inflammatory cell infiltrate.⁴⁶ The edema, in turn, contributes to hypovolemia and may promote increments in intracompartmental pressures in some muscle compartments, thus inducing further ischemia and tissue damage.

Bothrops venoms activate factor X and prothrombin, either alone or simultaneously.⁴⁵ They also have thrombin-like activity, converting fibrinogen into fibrin. These actions may lead to DIC. *Bothrops* venom may induce platelet function abnormalities, as well as a low platelet count.

Hemorrhagic manifestations result from the action of hemorrhagins, the metalloproteinases containing zinc described previously, in association with a low platelet count and coagulation abnormalities. Moreover, hemorrhagins are potent inhibitors of platelet aggregation.^{41,46}

Early and progressive pain and edema, bruises, blisters, and bleeding are frequently observed at the venom inoculation area. In the most severe cases, there is necrosis of soft tissues with abscess formation and the development of compartment syndrome, which may result in functional or anatomic loss of the affected limb.⁴¹ Systemic manifestations include bleeding (preexisting skin injuries, gingival bleeding, epistaxis, hematemesis, and hematuria), nausea, vomiting, excessive sweating, and hypotension. The most severe systemic complications are shock, AKI, septicemia, and a DIC-like syndrome.⁴¹

Kidney Involvement in *Bothrops* Bites

AKI associated with *Bothrops* envenomation has been attributed to hemodynamic changes, myoglobinuria, hemoglobinuria, coagulation abnormalities, and direct venom nephrotoxicity.⁴⁰ Myoglobinuria is unlikely to be an important factor in the pathogenesis of renal injury because *Bothrops* venom may cause localized muscular injury but does not have a systemic myotoxic effect like *Crotalus* venom and does not induce a significant increase in the creatine kinase level.⁴⁵

The development of hypotension or shock is a rare event after a *Bothrops* snakebite. Venom may cause hemodynamic abnormalities as a result of sequestration of fluids at the bite site, bleeding, and release of vasoactive substances. *Bothrops* venom is considered hemolytic in vitro, and there have been clinical reports of anemia and hemolysis after *Bothrops* envenomation, as well as reports of hemoglobinuria after the administration of *Bothrops* venom to rats. Hemoglobinuria might contribute to renal injury, worsening renal vasoconstriction, glomerular coagulation, and tubular nephrotoxicity.⁴⁷

Renal dysfunction after *Bothrops* poisoning occurs early, is usually severe and oliguric, and presents the need for dialysis in 33% to 75% of cases.⁴⁷ The most frequent renal structural injury found is acute tubular necrosis, although cases of bilateral cortical necrosis, interstitial nephritis, and acute glomerulonephritis with mesangial proliferation have also been reported.⁴⁷ The mortality rate of *Bothrops* venom-induced AKI ranges from 13% to 19%.⁴⁷

The management of snakebites involves immobilizing the injured part of the body to reduce the spread of venom and expediting transfer to an appropriate medical center. Tourniquets are not indicated because of the risk of significant ischemic damage. Expert consultation with a clinical toxicologist and/or poison control center is necessary for the timely administration of the appropriate antivenom.

CHRONIC KIDNEY DISEASE IN LATIN AMERICA

Latin America has undergone tremendous demographic and socioeconomic development in the past few decades. Improvement in sanitation, access to clean water, and expansion of health care programs have contributed to a significant reduction in infant mortality, as well as death due to malnutrition and infectious diseases, resulting in the prolongation of life expectancy. However, urbanization and assimilation of a Western lifestyle have resulted in an increase in noncommunicable diseases, such as diabetes, obesity, and hypertension, leading to a significant rise in kidney disease prevalence.^{48,49}

Currently, cardiovascular disease, diabetes, and malignancies are the leading causes of death in Latin America. Coping with this growth in noncommunicable diseases may be unsustainable and unaffordable for most countries, urging interventions to halt the trend. To achieve this goal, better characterization of the problem through data collection is critical. The Pan American Health Organization (PAHO) has developed strategies for the prevention and control of noncommunicable disease for the period 2012 to 2025, with a focus on cardiovascular diseases, cancer, diabetes and chronic respiratory diseases, their four common risk factors—tobacco use, unhealthy diet, physical inactivity, and harmful use of alcohol—and the related biologic risk factors such as hypertension and hyperglycemia.⁵⁰ Uruguay was the first country in Latin America to ban smoking in enclosed public spaces. Uruguay also has expanded access to primary health care services related to the early detection and risk factor prevention of noncommunicable diseases (NCDs). As a result, mortality linked to circulatory system diseases in the population 70 years of age and older has fallen from 75.18% in 2006 to 60.3% in 2014.^{51,52} Mexico, which has one of the highest prevalence rates for diabetes, overweight, and obesity in the world, has reduced the consumption of sugar-sweetened beverages through the implementation of an excise tax of 1 peso/L on these drinks since January 2014.⁵³

This section discusses the prevalence of, and risk factors and treatment strategies for, CKD in various Latin American countries, as well as the importance of the development of health programs for the earlier detection and prevention of CKD in this population.

PREVALENCE OF CHRONIC KIDNEY DISEASE

The prevalence of CKD in Latin America is difficult to assess owing to the lack of reliable and systematically recorded data. Most information comes from voluntary registries so it may be skewed or incomplete, leading to an inaccurate picture. In addition, these records are confounded by heterogeneity in the population screened and in the methods used to determine the glomerular filtration rate (GFR) and proteinuria. A recent report from Brazil on the prevalence of CKD based on Kidney Disease: Improving Global Outcomes (KDIGO) criteria has revealed a prevalence of 11.4% in a randomly screened population.⁵⁴ In a Mexican population-based study, the prevalence rate of creatinine clearance (C_{Cr}) less than 15 mL/min was 1142/million population and of C_{Cr} less than 60 mL/min was 80,788/million population.⁵⁵ Most of the data available have been based on advanced stages of CKD (end-stage kidney disease [ESKD]), showing that the prevalence has been growing by 7% annually during the past 10 years in Latin America. Nonetheless, the limited registry data of CKD in Latin America has revealed an average 12% prevalence of CKD compared with 11% in North America and 12% in Europe.^{56,57} CKD is a major cause of death in the region, ranking as high as second in Nicaragua and Mexico (Fig. 76.5)⁵⁸

From 1991 to 2013, the prevalence of ESKD in those undergoing RRT increased from 119 patients per million population (pmp) to 660 (hemodialysis [HD], 436 pmp; peritoneal dialysis [PD], 67 pmp; kidney transplantation, 157 pmp; Figs. 76.6 and 76.7). Only six countries have an RRT prevalence above the mean—Argentina, Brazil, Chile, Jalisco

	Data quality rating	1	2	3	4	5	6	7	8	9	10
Argentina	★★★★☆	IHD (0-66)	LRI (1-93)	Stroke (0-62)	Road Inj (0-75)	Lung C (0-84)	COPD (1-1)	Self-Harm (0-79)	Congenital (1-1)	NN Preterm (1-21)	CKD (1-72)
Chile	★★★★☆	IHD (0-35)	Stroke (0-62)	Road Inj (0-69)	Self-Harm (0-78)	Stomach C (1-47)	Congenital (1-05)	Lung C (0-47)	LRI (0-77)	CKD (1-59)	Alzheimer's (0-77)
Uruguay	★★★★☆	IHD (0-58)	Stroke (0-96)	Lung C (1-5)	Self-Harm (1-17)	Road Inj (0-76)	COPD (1-47)	LRI (1-14)	Colorect C (1-73)	Alzheimer's (1-39)	Breast C (1-42)
Colombia	★★★★☆	Violence (5-37)	IHD (0-45)	Road Inj (0-62)	Stroke (0-32)	Congenital (0-75)	LRI (0-55)	NN Preterm (0-66)	COPD (0-71)	CKD (1-09)	Self-Harm (0-47)
Costa Rica	★★★★★	IHD (0-41)	Road Inj (0-67)	CKD (1-79)	Congenital (0-78)	Violence (1-34)	Stroke (0-25)	Self-Harm (0-55)	Alzheimer's (1-12)	Stomach C (1-07)	NN Preterm (0-55)
El Salvador	★★★★☆	Violence (6-51)	IHD (0-63)	CKD (3-78)	Road Inj (0-78)	LRI (0-74)	Diabetes (1-5)	Alcohol (5-41)	Congenital (0-65)	Self-Harm (0-91)	Stroke (0-28)
Cuba	★★★★★	IHD (0-88)	Stroke (0-96)	Lung C (1-43)	LRI (1-27)	Self-Harm (0-75)	Alzheimer's (1-31)	COPD (1-01)	Colorect C (1-1)	CKD (1-73)	Prostate C (3-03)
Dominica	★★★★☆	IHD (0-57)	Diabetes (3-62)	Stroke (0-8)	CKD (3-37)	LRI (1-33)	Road Inj (0-81)	NN Enceph (3-73)	Prostate C (7-24)	Congenital (1-26)	HTN HD (3-24)
Dominican Republic	★★★★☆	IHD (0-67)	HIV (29-32)	NN Preterm (1-7)	Stroke (0-63)	Road Inj (0-82)	Violence (2-01)	Congenital (1-1)	LRI (0-89)	NN Sepsis (6-8)	Diabetes (0-96)
Puerto Rico	★★★★☆	IHD (0-6)	Diabetes (6-21)	Violence (7-75)	Stroke (0-59)	LRI (1-95)	CKD (3-53)	Road Inj (0-98)	Alzheimer's (0-99)	COPD (1-04)	Colorect C (0-79)
Guatemala	★★★★☆	LRI (1-13)	Violence (3-49)	IHD (0-38)	NN Preterm (0-58)	Diarrhea (0-6)	Road Inj (0-54)	Diabetes (1-47)	Congenital (0-73)	CKD (1-57)	Stroke (0-31)
Honduras	★★☆☆☆	IHD (0-86)	Violence (3-17)	Stroke (0-58)	NN Preterm (0-58)	CKD (1-68)	Road Inj (0-45)	Congenital (0-52)	LRI (0-25)	Diarrhea (0-7)	NN Enceph (0-62)
Mexico	★★★★☆	IHD (0-43)	CKD (4-23)	Diabetes (2-97)	Violence (2-97)	Road Inj (0-81)	Congenital (1-28)	LRI (0-88)	Stroke (0-32)	NN Preterm (2-91)	Cirr HepC (2-91)
Nicaragua	★★★★☆	IHD (0-41)	CKD (2-42)	LRI (0-39)	Road Inj (0-42)	Congenital (0-6)	Violence (1-03)	Diabetes (0-94)	Stroke (0-26)	NN Preterm (0-33)	Self-Harm (0-78)
Panama	★★★★☆	IHD (0-36)	Violence (3-32)	Road Inj (0-78)	Congenital (1-37)	LRI (1-18)	Stroke (0-45)	HIV (11-87)	Diabetes (1-57)	CKD (2-0)	NN Preterm (0-8)
Venezuela	★★★★☆	Violence (8-16)	IHD (0-67)	Road Inj (1-33)	Stroke (0-47)	Congenital (1-23)	NN Preterm (1-41)	CKD (2-26)	LRI (0-98)	Diabetes (1-62)	Self-Harm (0-77)
Bolivia	★☆☆☆☆	LRI (1-65)	NN Preterm (1-28)	IHD (0-54)	Congenital (1-25)	Road Inj (0-69)	Stroke (0-53)	NN Enceph (1-28)	CKD (1-66)	NN Sepsis (3-16)	Diabetes (0-95)
Ecuador	★★★★☆	Road Inj (0-98)	LRI (1-06)	IHD (0-32)	NN Preterm (1-09)	Violence (1-77)	Congenital (0-98)	CKD (1-92)	Stroke (0-35)	Diabetes (1-15)	Disaster (46-67)
Peru	★★★★☆	LRI (1-6)	IHD (0-77)	Road Inj (0-52)	Congenital (0-85)	NN Preterm (0-64)	F Body (5-97)	Stroke (0-26)	NN Enceph (1-18)	NN Sepsis (3-16)	CKD (1-08)
Brazil	★★★★☆	IHD (0-6)	Violence (4-73)	Road Inj (1-07)	Stroke (0-7)	LRI (1-13)	Congenital (0-9)	Diabetes (1-41)	COPD (0-97)	Alzheimer's (1-14)	NN Preterm (0-68)
Paraguay	★★★★☆	IHD (0-58)	Road Inj (1-05)	Stroke (0-68)	Violence (1-82)	Diabetes (1-71)	NN Preterm (0-97)	Congenital (1-0)	LRI (0-67)	CKD (1-72)	Self-Harm (0-57)

Fig. 76.5 Ranking of chronic kidney disease (CKD) as a cause of death in Latin America. CKD (highlighted in *green rectangle*) ranks within the top 10 causes of death in 18 of 20 countries and ranks among the top three causes of death in four countries. The Latin American rankings for CKD are the highest globally. (Modified from GBD 2016 Causes of Death Collaborators. Global, regional, and national age-sex specific mortality for 264 causes of death, 1980–2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet*. 2017;390(10100):1151–1210.)

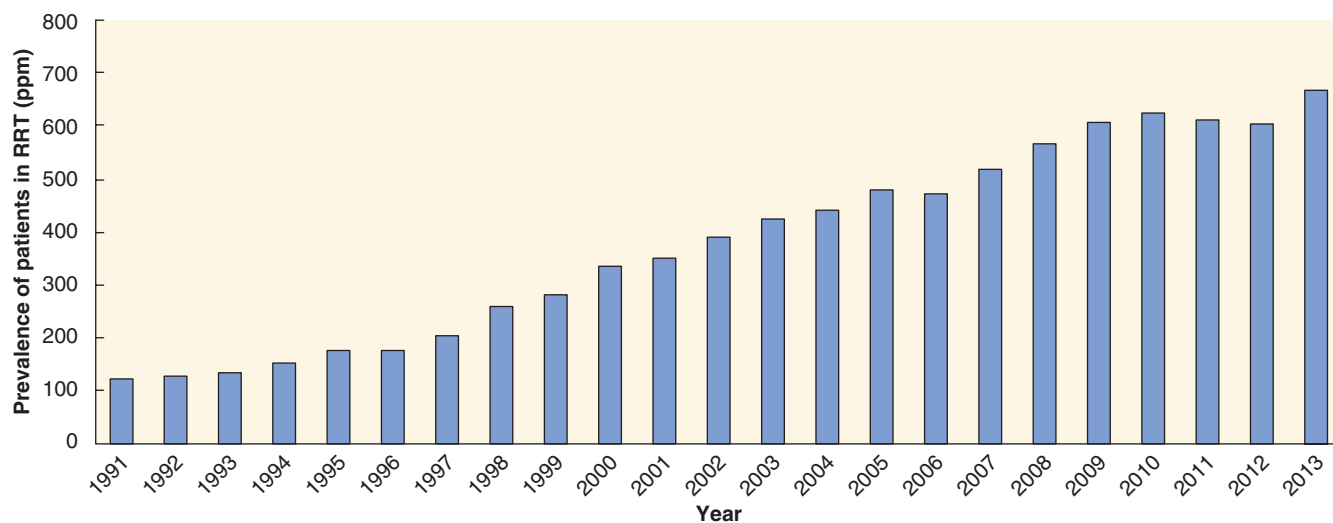


Fig. 76.6 Prevalence of renal replacement therapy over time in Latin America 1991–2013. RRT, Renal replacement therapy. (From Cusumano AM, Rosa-Diez GJ, Gonzalez-Bedat MC. Latin American Dialysis and Transplant Registry: experience and contributions to end-stage renal disease epidemiology. *World J Nephrol*. 2016;5(5):389–397.)

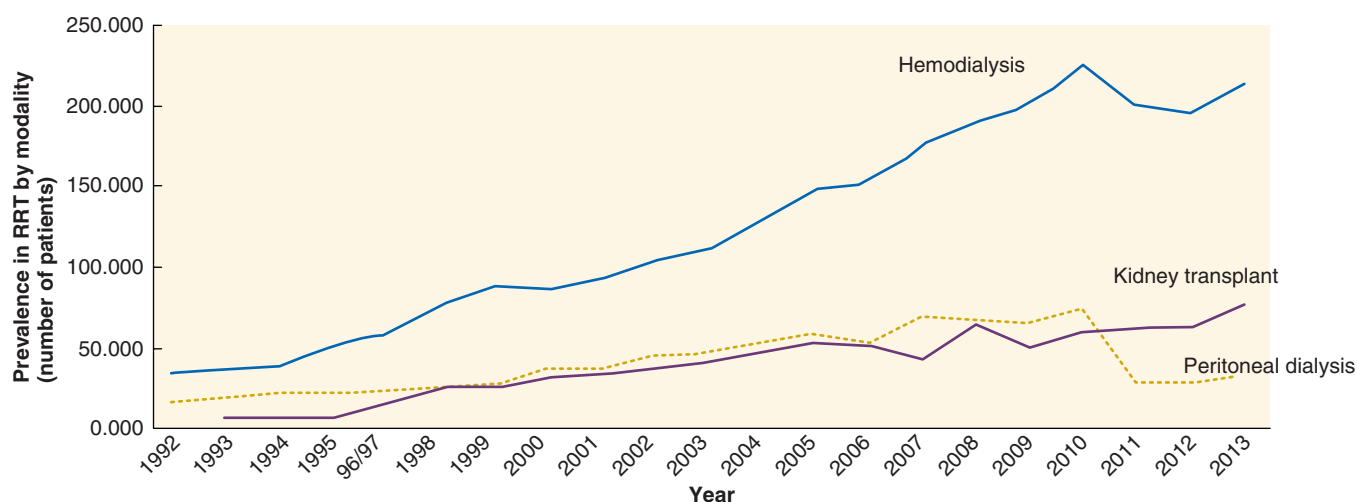


Fig. 76.7 Prevalence of different renal replacement therapy modalities in Latin America, 1991–2013. *RRT*, Renal replacement therapy. (Modified from Cusumano AM, Rosa-Diez GJ, Gonzalez-Bedat MC. Latin American Dialysis and Transplant Registry: experience and contributions to end-stage renal disease epidemiology. *World J Nephrol.* 2016;5(5):389–397.)

Table 76.1 Total Incidence Rates and Percentages of New Diabetic Renal Replacement Therapy Patients in Latin America

Country	Population	Incident Patients (pmp)	Incident Rate	Diabetics (%)
Argentina	4,220,935	6760	160.2	35.1
Bolivia	10,448,913	991	94.8	30.0
Brazil	202,740,000	36548	180.3	40.0
Chile	17,819,054	3230	182.4	16.7
Colombia	47,661,787	4274	89.7	33.5
Ecuador	161,00,000	2860	177.6	30.0
Guatemala	16,173,133	2018	124.8	30.0
Jalisco (Mexico)	7,742,303	3259	420.9	58.0
Nicaragua	6,146,000	150	24.4	41.6
Paraguay	6,783,374	137	20.2	45.3
Peru	30,297,279	910	30.0	32.2
Puerto Rico	3,615,000	1565	432.9	66.9
Uruguay	3,406,545	536	157.3	27.7

pmp, Per million population.

Modified from Cusumano AM, Rosa-Diez GJ, Gonzalez-Bedat MC. Latin American Dialysis and Transplant Registry: experience and contributions to end-stage renal disease epidemiology. *World J Nephrol.* 2016;5(5):389–397.

(Mexico), Puerto Rico, and Uruguay—with the reported rates ranging from 778 to 1847 pmp.⁵⁹

RISK FACTORS FOR CHRONIC KIDNEY DISEASE

Following the universal trend, the major risk factors for CKD in Latin America are hypertension and diabetes. Diabetic nephropathy is responsible for more than 30% of all cases of CKD in the region, although variation occurs (Mexico 40% vs. Argentina 20%) (Table 76.1).^{49,60} The epidemic of diabetes is expected to continue worldwide, and an increase of more than 248% is forecast for the region from 2000 to 2030. Hypertension also is a silent cause of CKD, with a prevalence as high as 30% in the Mexican population. An evaluation of 8883 individuals in Southern Brazil, self-referred from the general population during a health fair, has shown

that 16% of the population screened has a history of hypertension.⁶¹ A family history of hypertension or diabetes was present in 42%. In addition, 30% of patients evaluated in this screening program had signs of hematuria, 6% had proteinuria, and 3% had hematuria and proteinuria.⁶¹ In a Mexican national screening study of the general population, 9.2% of participants had proteinuria, which was strongly associated with hypertension, diabetes, obesity, and age.⁶² Sedentary lifestyle, smoking, and unhealthy dietary habits, which included high-fat and high-carbohydrate consumption in Latin America, have contributed to the described comorbidities and expanding incidence of obesity.

Other important risk factors for CKD in the region include infectious complications, glomerulonephritis, and interstitial nephritis of unclear cause (discussed later in “Mesoamerican Nephropathy”).

In summary, it is clear that the rising incidence of CKD is a major public health concern in Latin America, and diligent assessment of its prevalence, geographic distribution, and risk factors will be critical to establishing appropriate prevention programs for early screening and intervention. Uruguay has initiated a prevention program enrolling patients with CKD, and preliminary results have shown a significant improvement in blood pressure and glycemic control in association with a reduction in the rate of deterioration of renal function—from 3.1 mL/min per year to −0.13 mL/min per year in patients with CKD stages 2 to 3.⁶³ In Southern Brazil, the Pro-Renal Foundation has taken a leading role in raising awareness of CKD in the community.⁶¹ It has also been a local pioneer in providing multidisciplinary care to patients with CKD with teams of nephrologists, nurses, social workers, dentists, podiatrists, psychologists, and dietitians.⁶⁴

These successes demonstrate that it is possible to modify health trends in Latin America with the help of the local community. Even basic interventions, such as encouraging lifestyle changes through exercise and dietary substitutions, in combination with augmenting the awareness of blood pressure and blood glucose control, may yield significant reductions in incidence and progression of kidney disease.

MESOAMERICAN NEPHROPATHY—CHRONIC KIDNEY DISEASE OF UNKNOWN CAUSE

The World Health Organization has demonstrated a high prevalence of CKD in regions of Nicaragua, Costa Rica, and El Salvador, with the highest mortality in El Salvador.⁶⁵

Over the last decade, numerous reports have emerged from Central America regarding a rising prevalence of CKD, particularly among agricultural workers. CKD in this region seems to have a higher prevalence in men, favoring regions at lower altitudes, and particularly affecting workers in sugar cane plantations.^{66–71}

These early reports described a clinical presentation characterized by an elevation of serum creatinine level and nonnephrotic proteinuria in young men without evidence of any known risk factors for CKD such as hypertension or diabetes, raising the possibility of a tubulointerstitial kidney disease due to an unknown environmental or occupational hazard.^{72–74} Another hypothesis was an ischemic injury as a result of the heavy workload in a hot climate leading to chronic dehydration and hypovolemia. The significance of heat stress was suggested by the male preponderance and high prevalence in low-altitude farming villages close to the Pacific Ocean, where it is hotter. The combination of repeated episodes of dehydration during heavy manual work, and perhaps the concomitant use of NSAIDs, could explain the renal lesion. Nonetheless, it is clear that a second “hit” would be necessary for significant kidney injury to develop in such healthy young men. This endemic form of chronic kidney disease of unknown cause was named “Mesoamerican nephropathy” (MeN) and seemed to be endemic to Pacific coastal communities in Central America.⁶⁹

Wijkström and colleagues presented the first study on the correlation of clinical data with biochemical and morphologic findings in patients with MeN. The investigators analyzed renal biopsy samples from eight male sugarcane workers (aged 22–57 years) from a rural area of El Salvador who

Table 76.2 Potential Causes of Mesoamerican Nephropathy

Cause	Examples
Toxins	Pesticides and agricultural chemicals, heavy metals, herbal toxins
Drugs	Nonsteroidal antiinflammatory drugs
Infections	Leptospirosis, pyelonephritis
Genetic, developmental susceptibility	Low birth weight, specific genetic predisposition of unclear origin (APOL1-like)
Heat-related volume depletion	Ischemia, repeated rhabdomyolysis, fructose-mediated injury

had presented with impaired renal function (estimated GFR, 27–77 mL/min/1.73 m²), inactive urinary sediment, and nonnephrotic proteinuria.^{66,75} The morphologic picture was similar in biopsy specimens of all included patients. Chronic tubulointerstitial damage with tubular atrophy and interstitial fibrosis was found, in combination with chronic glomerular changes. The most striking and surprising finding was the presence of relatively extensive global sclerosis, involving 29% to 78% of glomeruli analyzed.⁶⁶ These biopsies confirmed the suspicion of a primary tubulointerstitial disease. Immunofluorescence and electron microscopy did not show immune complex deposits, so an immune complex disease seemed unlikely.⁶⁶

In one study, recurrent dehydration has been shown to raise serum osmolality and activate the polyol pathway in the kidney, leading to increased conversion of glucose to sorbitol and fructose. Fructose in turn can be metabolized by the fructokinase that is constitutively present, resulting in the generation of uric acid, oxidants, and chemokines, which can cause local tubulointerstitial injury.^{76,77} The same group has also shown that repeated dehydration can cause chronic kidney injury in mice and that this injury is fructokinase-dependent, because mice with knockout of fructokinase are protected.⁷⁶ Thus, it is possible that MeN may be a kidney disease due to chronic dehydration and that recurrent dehydration is almost certainly a component of its pathogenesis, leading to tubulointerstitial injury.^{78,79} However, toxins (e.g., pesticides) or heavy metals might contribute as a second factor in the development of the disease. Although originally reported in Central America, MeN could represent a larger entity, explaining CKD epidemics seen in countries such as Sri Lanka, Bangladesh, and central Australia (Table 76.2; see related discussion in Chapters 75, 77, and 79).⁶⁸

END-STAGE KIDNEY DISEASE IN LATIN AMERICA

The prevalence of ESKD in Latin America is extrapolated from data provided about RRT in the region. However, due to the heterogeneous availability of RRT according to the nations’ health care coverage and budget, these numbers do not clearly represent the true prevalence. The best available information comes from the Latin American Dialysis and

Kidney Transplant Registry (LADKTR), a committee of the Latin American Society of Nephrology and Hypertension (SLANH), that has been collecting data since 1991 (Table 76.3). These data show a growth of more than 200% in the prevalence of ESKD in Latin America over the past 13 years (see Fig. 76.6).

Although RRT is provided to all patients with a diagnosis of ESKD in Argentina, Brazil, Chile, Cuba, Puerto Rico, Uruguay, and Venezuela, some countries may provide RRT for as few as 20% of these patients and other countries, such as Bolivia, may pay for only twice-weekly dialysis. Furthermore, Latin American middle-income countries have a disproportionately different prevalence of RRT, which seems to correlate with the type of health coverage provided. Universal RRT coverage is associated with the highest prevalence of RRT (e.g., Argentina, Brazil) compared with no universal access (e.g., Nicaragua, Guatemala, Paraguay).⁸¹ The consequence of this heterogeneous coverage is that the prevalence rate of RRT may vary enormously among countries (see Table 76.3). Thus, the prevalence rate is more than 600 pmp in Puerto Rico, Chile, Uruguay, and Argentina; between 300 and 600 pmp in Colombia, Brazil, Mexico, Panama, and Venezuela; and significantly lower, perhaps less than 100 pmp, in the remaining countries.⁸¹ The leading cause of ESKD in Latin America is diabetes, with the highest incidences reported in Puerto Rico (66.9%), Mexico (58.0%), Paraguay (45.3%), and Brazil (40.0%).⁵⁶

The low prevalence of ESKD in some regions should raise the suspicion of underdiagnosis and/or lack of available resources to offer RRT. The direct correlation of the degree of development of a country with the prevalence of RRT suggests that access to ESKD care is an important component. One report has confirmed this relationship by showing that RRT prevalence and kidney transplantation rates correlate significantly with gross domestic product, health expenditure, proportion of the population older than 65 years, life expectancy at birth, and proportion of the population living in urban settings.⁸² Owing to the substantial economic burden of providing dialysis in many countries that still lack essentials such as sanitation, prevention strategies to halt kidney disease progression are critical.

TRANSPLANTATION

Renal transplantation has also developed significantly in Latin America. The data from the LADKTR indicated that the overall kidney transplantation rate increased from 3.7 pmp in 1987 to 6.9 in 1991, and then to 19.4 in 2013, although, there were remarkable disparities among the various countries in the last year—57.7 pmp in Jalisco (Mexico), 32 pmp in Uruguay, and 1.8 pmp in Nicaragua. The highest number of transplants ($n = 5433$) was in Brazil, which had a transplantation rate of 26.8 pmp for 2013.⁵⁶ More recent data from Brazil have revealed an annual total transplant number of 5648 in 2015 (27.9 pmp), followed by Mexico, with 2707, Argentina, 1198, and Colombia, 761. In 2016, Brazil had performed 5492 kidney transplantations—4292 deceased donors, 1200 living donors (171 partners, 80 unrelated).⁸³ These rates are still remarkably lower than those in other developed countries, such as the United States (46 pmp) and Sweden (41 pmp). About 52% of these transplants are

from deceased donors, though some countries, such as Nicaragua, Peru and El Salvador, performed solely living donor kidney transplantations.

Brazil has been one of the countries with the greatest expansion in kidney transplantation numbers, in part because of its free universal health coverage system, which includes full ambulatory and hospital medical care, along with provision of all immunosuppressive drugs for patients undergoing transplantation. A single city in Brazil (São Paulo) has developed the largest kidney transplantation program in the world.⁸³ In addition to increased efficiency in organ procurement, several other factors have contributed to this expansion, including the development of a dedicated hospital (Hospital do Rim) for kidney transplantations and a system for rapid workup of recipients and donors.⁸³ This program has approximately 5011 patients on the waiting list, and the recipient selection is based on human leukocyte antigen (HLA) matching. A significant proportion of the recipients are black and have been on dialysis for a long time. More than 700 first consultations for living donation are done every year, and most patients are followed locally after transplantation, with more than 200 appointments/day. Transplantation outcomes among living donor recipients are comparable to those reported in large registries but inferior outcomes have been observed among recipients of deceased donor organs. This is likely due to prolonged ischemia time due to poor care of critically ill potential deceased donors and logistic difficulties from a lack of financial support in organ allocation.^{83,84}

The criteria for brain death have been well established and culturally accepted in all Latin American countries, and family consent is required for organ donation.⁸⁵ The regulations for living donors are also well defined, with restrictions against unrelated donors and prohibition of any kind of organ commerce. Access to transplantation is limited by the health care models of individual countries. In addition, lack of an organized organ procurement system leads to significant limitations in the number of organs available in many countries. One of the greatest challenges is to reduce the regional disparities related to access to transplantation. Furthermore, improvements and resources could be better used, with the development of a Latin American registry of outcomes of transplanted patients and living donors. Finally, clinical and experimental studies to understand the transplantation-related immune response of the ethnically diverse Latin American population better would be critical to advance the transplantation field in this region further.

TRENDS AND THE FUTURE

The greatest challenge for the nephrology community in Latin America in the coming years will be dealing with the growing number of patients with ESKD, the lack or incompleteness of health care coverage for RRT in some countries, late referrals, shortage of trained personnel, small kidney transplantation programs, and lack of registries. In addition, the early identification and aggressive treatment of infectious diseases complicated by AKI may minimize long-term renal sequelae. In spite of the worldwide concern about the increasing number of patients with ESKD and efforts to identify risk groups, as well as the development of strategies for the

Table 76.3 Socioeconomic Indexes, Prevalence and Incidence of Renal Replacement Therapy and Transplantation Rate in Latin America

Country	Population	GNI	LEB	HD	PD	Prevalence Rates, pmp			Incidence Rate	Tx by			Kidney Tx, n	Kidney Tx, Rate	Nephrologists, n	Nephrologists, pmp
						Total Dialysis	Lfg	Total RRT		Deceased Donors, %	Kidney Tx, n	Kidney Tx, Rate				
Argentina	42,202,935	13,690	76	626.6	36.0	662.7	197.2	859.9	160.2	68.4	1287	30.5	1150	27.2		
Bolivia	10,448,913	2220	67	195.2	18.3	213.5	31.6	245.1	94.8	24.0	75	7.2	24	2.3		
Brazil	202,740,000	11,640	74	449.6	45.6	495.2	212.6	707.8	180.3	74.7	5433	26.8	3300	16.3		
Chile	17,819,054	14,290	80	1019.1	61.2	1080.3	205.1	1285.4	182.4	74.8	234	13.1	132	7.4		
Colombia	47,661,787	7020	74	349.0	143.6	492.6	111.3	603.9	89.7	99.7	680	14.3	95	2.0		
Costa Rica	4,773,730	8850	80	42.3	76.0	118.4	282.6	400.9	ND	48.6	105	22.0	24	5.0		
Cuba	11,163,934	6051	79	259.1	10.1	269.3	78.4	347.6	103.1	ND	174	15.6	524	46.9		
Ecuador	16,100,000	3600	76	481.8	48.0	529.8	20.4	550.2	177.6	81.1	127	7.9	143	8.9		
El Salvador	6,401,240	5360	72	232.5	288.7	521.1	73.6	594.7	390.1	0.0	20	3.1	47	7.3		
Guatemala	16,173,133	3130	72	157.7	221.3	379.0	54.0	433.0	124.8	13.3	90	5.6	54	3.3		
Honduras	8,500,000	2140	73	186.9	14.4	201.3	8.4	209.6	176.7	0.0	0	0.0	18	2.1		
Jalisco (Mexico)	7,742,303	ND	ND	599.4	486.7	1086.1	567.4	1653.5	420.9	16.1	447	57.7	45	5.8		
Nicaragua	6,146,000	1690	74	211.5	24.4	235.9	21.2	257.1	24.4	0.0	11	1.8	28	4.6		
Panamá	3,975,404	9030	77	495.0	90.3	585.3	110.7	696.0	462.1	73.1	48	12.1	25	6.3		
Paraguay	6,783,374	3310	75	165.7	4.0	169.7	19.9	189.6	20.2	79.0	26	3.8	46	6.8		
Perú	30,297,279	5680	72	272.2	43.1	315.3	63.2	378.5	30.0	75.0	184	6.1	301	9.9		
Puerto Rico	3,615,000	18,370	79	1362.1	106.2	1468.3	378.4	1846.7	432.9	86.3	80	22.1	97	26.8		
Rep Dominicana	12,000,000	5570	73	178.8	47.3	226.1	52.8	278.9	208.3	92.9	84	7.0	135	11.3		
Uruguay	3,406,545	13,670	77	692.2	71.6	763.8	323.5	1087.3	157.3	91.4	105	30.8	173	50.8		
Venezuela	30,389,596	12,460	74	505.1	0.0	505.1	60.8	565.9	ND	69.8	281	9.2	502	16.5		
Totals	488,340,227	147,771	75	442.0	67.0	509.0	159.0	669.0	149	70.4	9491	19.4	6863	14.0		

prevention of CKD, there is little activity in many Latin American countries. Dialogue between the nephrology community and government authorities in the region has to be based on data, which are lacking. Only a few countries have reliable registries of ESKD and transplantation programs. The lack of universal health care coverage in some countries and, in many cases, a restrictive renal replacement program, poses a much greater challenge because it reflects the public health agenda of a particular country and is a difficult scenario to modify. It is to be hoped that the combined efforts of scientific, nephrology and transplantation societies, kidney foundations, and patient organizations will advocate for and implement screening and prevention programs for CKD, educate the community, establish registries and more active kidney transplantation programs, and engage policy makers regarding the importance of kidney disease prevention and treatment.



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BOARD REVIEW QUESTIONS

1. A 35-year-old male agricultural worker from Central America comes for a routine physical checkup and was found to have an elevated creatinine level (2.2 mg/dL; it was 1.5 mg/dL 1 year prior). Urinalysis reveals no proteinuria, hematuria, or leukocyturia. He is otherwise healthy and denies hypertension, diabetes, and smoking. His family history is remarkable for prostate cancer. The most likely finding on his kidney biopsy is:
 - a. Tubulointerstitial disease
 - b. Acute tubular injury
 - c. Glomerulonephritis
 - d. Necrotizing granuloma
 - e. Thrombotic microangiopathy

Answer: a

Rationale: Numerous reports from Central America have indicated a rising prevalence of chronic kidney disease, particularly among agricultural workers. The first study on the correlation of clinical data with biochemical and morphologic findings in patients with Mesoamerican nephropathy was reported. This was based on analyses of renal biopsy samples from eight male sugar cane workers (aged 22–57 years) from a rural area of El Salvador who had presented with impaired renal function (estimated glomerular filtration rate, 27–77 mL/min/1.73 m²), inactive urinary sediment, and nonnephrotic proteinuria. The morphologic picture was similar in biopsy specimens of all included patients. Chronic tubulointerstitial damage with tubular atrophy and interstitial fibrosis was found in combination with chronic glomerular changes. The most striking and surprising finding was the presence of relatively extensive global sclerosis, involving 29% to 78% of glomeruli analyzed. These biopsies confirmed the suspicion of a primary tubulointerstitial disease. Immunofluorescence and electron microscopy did not show immune complex deposits, so an immune complex disease seems unlikely.

2. A 25-year-old woman has returned from a trip to Brazil 10 days prior and now presents with fever, headache, and maculopapular rash for 3 days. Laboratory results revealed an elevated creatinine level (1.5 mg/dL from baseline, 1 mg/dL) and urine sediment with dysmorphic red blood cells. Based on her recent trip to an endemic area of dengue fever, acute dengue infection is suspected. What is the ideal treatment for dengue?
 - a. Antiviral therapy with valganciclovir
 - b. Antidengue antibody
 - c. Supportive care
 - d. Antibacterial therapy with doxycycline
 - e. Intravenous immunoglobulin

- a. Antiviral therapy with valganciclovir
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- e. Intravenous immunoglobulin

Answer: c

Rationale: There is no specific treatment for dengue fever, so management consists of supportive care, including controlling fever (acetaminophen), preventing dehydration (oral [PO] or intravenous [IV] fluids), and managing complications such as bleeding.

3. A 55-year-old woman with a history of kidney transplantation is planning a trip to an endemic area of yellow fever. Her graft function has been stable for the past 3 years (creatinine, 1.3 mg/dL) and she is currently on tacrolimus and mycophenolate mofetil. What is the best next step to prevent yellow fever infection during her trip?
 - a. Yellow fever vaccination
 - b. Covering exposed skin areas and using mosquito repellent
 - c. Prophylactic antimicrobial
 - d. High-dose vitamin C

Answer: b

Rationale: Yellow fever (YF) is a viral hemorrhagic fever transmitted by infected mosquitoes. One of the most important mechanisms for preventing yellow fever is vaccination. The live attenuated vaccine against YF yields protective immunity in about 90% of individuals within 10 days of receiving the 0.5-mL subcutaneous dose and in nearly 100% of individuals within 3 to 4 weeks after vaccination. A booster dose is recommended every 10 years. The World Health Organization (WHO) recommends vaccination for travelers to YF-endemic areas of Africa and South America, as well as for residents of those areas. However, the vaccine is contraindicated in persons who have known egg allergy or are immunosuppressed (e.g., transplant recipients). Furthermore, there are no effective treatments against YF other than supportive care. Therefore the best next step to prevent YF is to cover exposed skin areas and use mosquito repellent to prevent *Aedes aegypti* bite.